

University of Arkansas STAT6393V - Week 3

Week 3 - Assignment 3: Model Comparison Australia Electricity Demand

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Introduction

Basics and setup

The goal of this assignment is to compare three different models for estimating the median and 95th percentile of electricity consumption:

1. The Normal distribution model from Assignment 2 (reuse results from Assignment 2)
2. A quantile regression (QR) model which simultaneously estimates both quantiles
3. The estimate of the quantiles from SPQR

The data processing pipeline is identical from Assignment 2.

```
rm(list = ls())
data("AUDem", package = "qgam")
meanDem <- AUDem$meanDem
library(lubridate) # need this for date-time tinkering
months <- month(meanDem$date) #might throw warnings, can be ignored
years <- year(meanDem$date) #might throw warnings, can be ignored
Y <- meanDem$dem
```

The quantile regression loss function is:

```
library(torch); library(cito); library(SPQR)
q2 = torch_tensor(0.5)
q3 = torch_tensor(0.95)
loss_func = function(pred, true,...) {
  l2 = torch_max(q2*(true[,1,drop=FALSE]-pred[,1,drop=FALSE]),
    other = (1.0-q2)*(pred[,1,drop=FALSE]-true[,1,drop=FALSE]))
  l3 = torch_max(q3*(true[,2,drop=FALSE]-pred[,2,drop=FALSE]),
    other = (1.0-q3)*(pred[,2,drop=FALSE]-true[,2,drop=FALSE]))
  return(l2+l3)
}
```

The QR loss is the same one from last week, but I dropped the lowest quantile from it. Covariate cleanup as before:

```
X <- meanDem[,c(2,5,7)]
X <- cbind(X,months,years)
dem_df <- cbind(Y,X)
```

Question 1 - Fitting models

Fit $Y \sim .$ using QR and SPQR defined above. You should be able to scrounge together code for both models from what we covered in class. Guidelines for each model:

1. Have *at least* 0.2 validation split in each case
2. No need to have a separate test set
3. For QR - use the training/validation loss as the metric for choosing your final model
4. For SPQR - you can use the loss as the metric, but you can also use the `plotGOF` function to assess goodness of fit.

Example for SPQR

```
library(SPQR)
control <- list(epochs = 300, lr = 0.001, batch.size = 100)
spqr_fit <- SPQR(X = X, Y = Y, method = 'MLE', normalize = T, n.hidden =
  ↪ c(50,50),
               activation = 'relu', verbose = F, control = control, seed =
  ↪ 303)
#summary(spqr_fit)
#plotGOF(spqr_fit)
```

Question 2 - Getting predictions

Do the following steps for each of your models:

1. Subset the data for December. Let's just compare the summer data.
2. Get predictions for each model for each season. There should therefore be 2 different prediction dataframes. Save them as `qr_12` and `spqr_12`.

Question 3 - Getting quantiles

The way you get quantiles for each model will be different. I will give the broad outline of the process below for both models.

• QR

You should have 2 columns in the predictions for each month - the first column will correspond to the median, the second will correspond to the 0.95 quantile. You can save your eventual outputs as `qr_12_50`, `qr_12_95`.

• SPQR

The predict function will give you a dataframe with 2 columns, just like for QR

```
spqr_pred <- predict(spqr_fit, X = as.matrix(X), type = 'QF', tau =
  ↪ c(0.50,0.95))
dim(spqr_pred)
```

Note how I needed to use `as.matrix` in the predict statement but not in the `fit` statement. That's a bug, not a feature. Separate out the 2 columns in 2 different vectors - `spqr_12_50`, `spqr_12_95`.

Question 4 - Model comparison

Import in your predictions from Assignment 2. You can either do it using the `load` function, or by going to the environment tab and use the “load workspace” option.

We will do a total of 6 tests (3 methods x 2 quantile levels). Use `wilcox.test` as before with the following options

1. `x` and `y` vectors
2. `paired = T`
3. `alternative = 'two.sided'`

What should be in the final submission?

Report the:

1. Final architectures you chose for each model
2. A table for the median, and a table for the 0.95 quantile. Each table should have the form:

	Gaussian	QR	SPQR
Gaussian	X		
QR		X	
SPQR			X

The diagonals will be empty. The off diagonals will have the p-values from the tests.

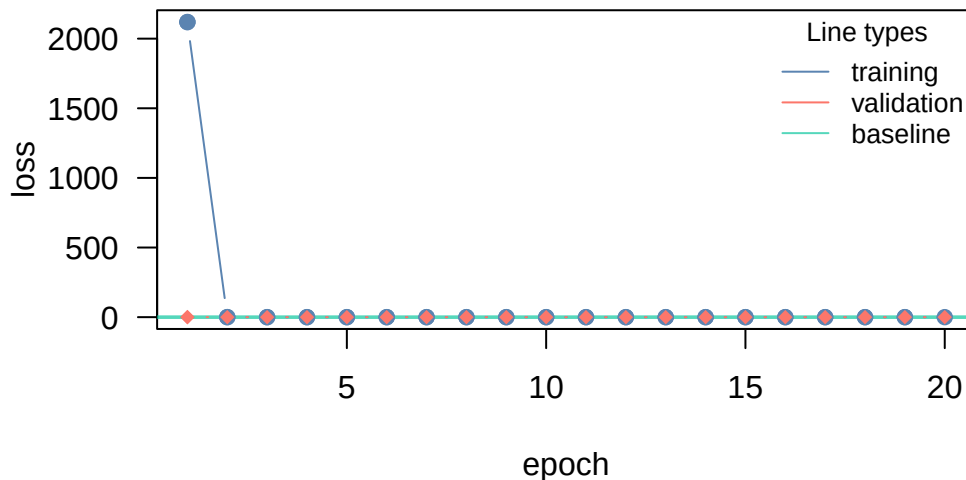
i Answer 1 - Fitting models

Quantile Regression with Train/Validation Loss. Denoted as Model 2 (m2)

```
m2 <- cito::dnn(cbind(Y,Y)~., data = dem_df,
  lr = 0.01,
  validation = 0.2,
  loss = loss_func,
  lambda = 0.000, alpha = 0.5,
  epochs = 20L, hidden = c(30L, 30L),
  activation = "relu", verbose = TRUE, plot = TRUE)
```

Loss at epoch 1: training: 2119.258, validation: 0.344, lr: 0.01000

Training of DNN



Loss at epoch 2: training: 0.215, validation: 0.126, lr: 0.01000
Loss at epoch 3: training: 0.088, validation: 0.064, lr: 0.01000
Loss at epoch 4: training: 0.061, validation: 0.062, lr: 0.01000
Loss at epoch 5: training: 0.060, validation: 0.061, lr: 0.01000
Loss at epoch 6: training: 0.059, validation: 0.060, lr: 0.01000
Loss at epoch 7: training: 0.058, validation: 0.059, lr: 0.01000
Loss at epoch 8: training: 0.057, validation: 0.058, lr: 0.01000
Loss at epoch 9: training: 0.057, validation: 0.057, lr: 0.01000
Loss at epoch 10: training: 0.056, validation: 0.056, lr: 0.01000
Loss at epoch 11: training: 0.055, validation: 0.055, lr: 0.01000
Loss at epoch 12: training: 0.054, validation: 0.055, lr: 0.01000
Loss at epoch 13: training: 0.053, validation: 0.054, lr: 0.01000
Loss at epoch 14: training: 0.052, validation: 0.053, lr: 0.01000
Loss at epoch 15: training: 0.052, validation: 0.052, lr: 0.01000
Loss at epoch 16: training: 0.051, validation: 0.052, lr: 0.01000
Loss at epoch 17: training: 0.050, validation: 0.051, lr: 0.01000
Loss at epoch 18: training: 0.049, validation: 0.050, lr: 0.01000
Loss at epoch 19: training: 0.049, validation: 0.050, lr: 0.01000
Loss at epoch 20: training: 0.048, validation: 0.049, lr: 0.01000

The model's loss falls steeply within as few as the first 10 epochs, so it stands to reason that we do not need that many epochs.

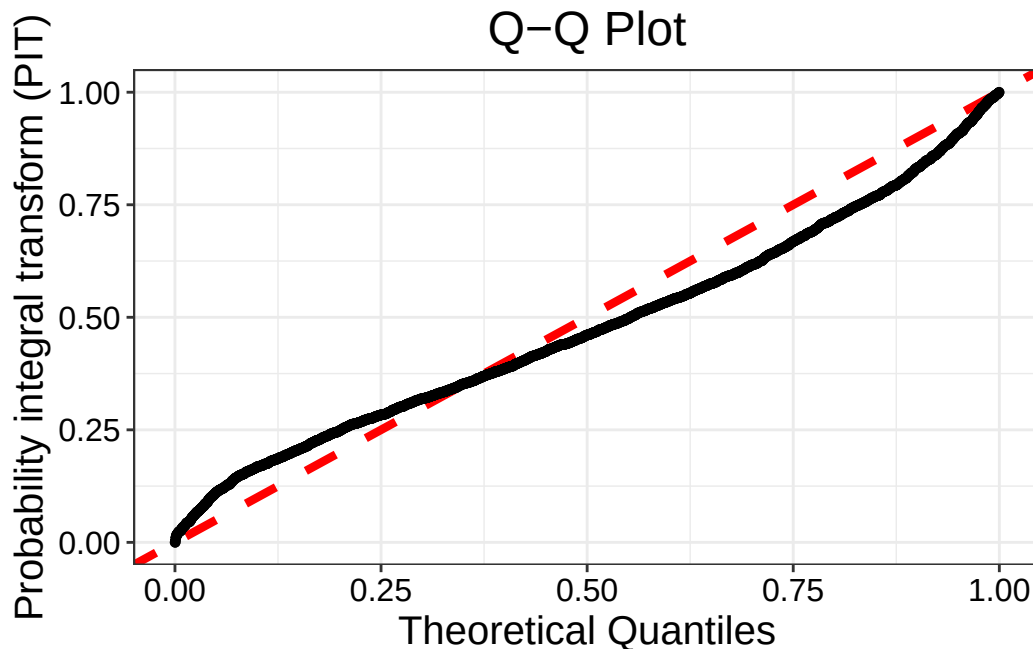
Using SPQR, denoted as model 3 (m3):

```
control <- list(epochs = 300, lr = 0.001, batch.size = 100, valid.pct = 0.2)
m3 <- SPQR(X = X, Y = Y, method = 'MLE', normalize = T, n.hidden = c(50,50),
activation = 'relu', verbose = F, control = control, seed = 303)
```

Assessing SPQR model's goodness of fit:

```
plotGOF(m3)
```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
i The deprecated feature was likely used in the SPQR package.
Please report the issue at <<https://github.com/stevengxu/SPQR/issues>>.



The shape of the QQ-plot suggests that the model fit the data reasonably well, in spite of a small “S” shape in the data points.

i Answer 2 - Getting Predictions

Predictions:

```
dem12 <- dem_df[months == 12,]
```

```
qr_12 <- predict(m2,dem12[,-1]) # QR approach
```

```
spqr_12 <- predict(m3, X = as.matrix(dem12[,-1]), type = "QF", tau =  
  ↪ c(0.50,0.95)) # SPQR approach
```

i Answer 3 - Getting Quantiles

50% and 95% quantiles for QR:

```
qr_12_50 <- qr_12[,1]
qr_12_95 <- qr_12[,2]
```

50% and 95% quantiles for SPQR:

```
spqr_12_50 <- spqr_12[,1]
spqr_12_95 <- spqr_12[,2]
```

i Answer 4 - Model Comparison

Note: Loading Gaussian model from Assignment 2 (I will load the whole model specification, but the predictions should be saved on the second assignment and then imported here as an object).

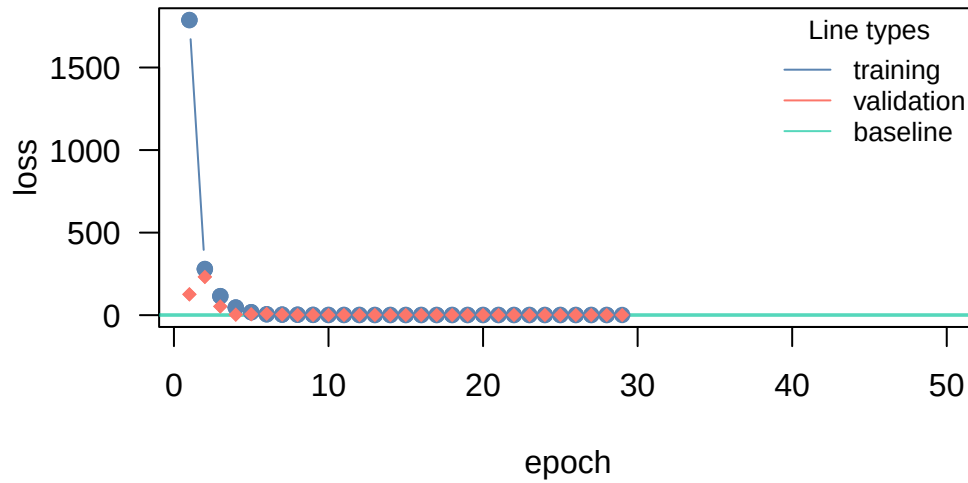
Importing the results from Assignment 2 as objects:

```
# load the results from Assignment 2
load('Assignment2_results.RData')

# Model 1 loss function
gaussian_nll = function(pred, true, ...) {
  loss = -torch::distr_normal(pred, scale =
  ↪ torch::torch_exp(scale_par))$log_prob(true)
  return(loss$mean())
}

m1 <- cito::dnn(
  Y ~ ., data = dem_df,
  loss = gaussian_nll,
  epochs = 50,
  verbose = F,
  validation = 0.2,
  custom_parameters = list(scale_par = 0.0),
  activation = "relu",
  optimizer = 'adam'
)
```

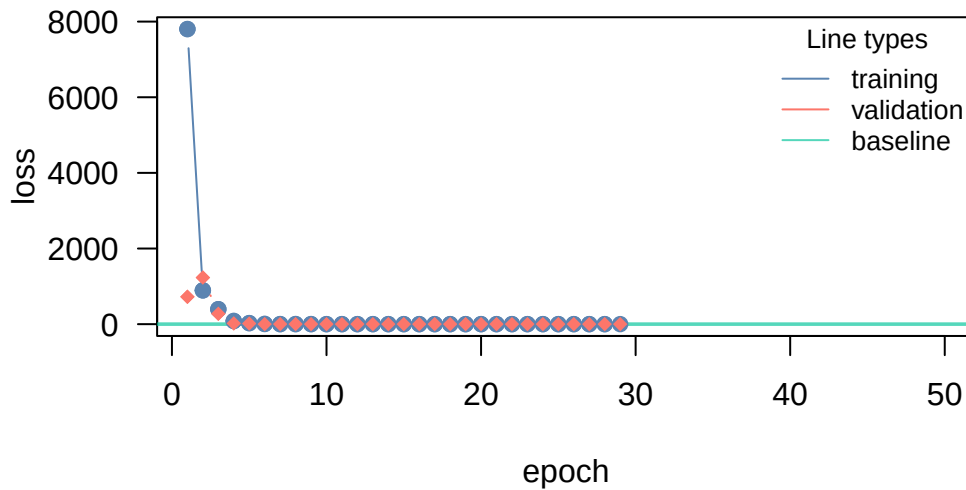
Training of DNN



```
gaussian_12 <- predict(m1, dem12[, -1])  
gaussian_12_50 <- c(gaussian_12)
```

```
## Model 1 loss function  
gaussian_nll = function(pred, true, ...) {  
  loss = -torch::distr_normal(pred, scale =  
    ↪ torch::torch_exp(scale_par))$log_prob(true)  
  return(loss$mean())  
}  
m1 <- cito::dnn(  
  Y ~ ., data = dem_df,  
  loss = gaussian_nll,  
  epochs = 50,  
  verbose = F,  
  validation = 0.2,  
  custom_parameters = list(scale_par = 0.0),  
  activation = "relu",  
  optimizer = 'adam'  
)
```

Training of DNN



```
gaussian_12 <- predict(m1, dem12[, -1])

gaussian_12_50 <- c(gaussian_12)

# Generating 0.95 quantile predictions with the estimated
# Gaussian parameters from the model
gaussian_12_95 <- c(qnorm(0.95, mean = gaussian_12, sd =
  ↪ exp(m1$parameter$scale_par)))
```

Model comparisons:

```
# Gaussian - QR Dec 50
wilcox.test(x = gaussian_12_50, y = qr_12_50, alternative = 'two.sided',
  ↪ paired = T)
```

Wilcoxon signed rank test with continuity correction

```
data: gaussian_12_50 and qr_12_50
V = 39059, p-value < 2.2e-16
alternative hypothesis: true location shift is not equal to 0
```

```
# Gaussian - QR Dec 95
wilcox.test(x = gaussian_12_95, y = qr_12_95, alternative = 'two.sided',
  ↪ paired = T)
```

Wilcoxon signed rank test with continuity correction

data: gaussian_12_95 and qr_12_95
V = 39060, p-value < 2.2e-16
alternative hypothesis: true location shift is not equal to 0

```
# Gaussian - SPQR Dec 50  
wilcox.test(x = gaussian_12_50, y = spqr_12_50, alternative = 'two.sided',  
  ↪ paired = T)
```

Wilcoxon signed rank test with continuity correction

data: gaussian_12_50 and spqr_12_50
V = 39003, p-value < 2.2e-16
alternative hypothesis: true location shift is not equal to 0

```
# Gaussian - SPQR Dec 95  
wilcox.test(x = gaussian_12_95, y = spqr_12_95, alternative = 'two.sided',  
  ↪ paired = T)
```

Wilcoxon signed rank test with continuity correction

data: gaussian_12_95 and spqr_12_95
V = 39060, p-value < 2.2e-16
alternative hypothesis: true location shift is not equal to 0

```
# QR - SPQR Dec 50  
wilcox.test(x = qr_12_50, y = spqr_12_50, alternative = 'two.sided', paired  
  ↪ = T)
```

Wilcoxon signed rank test with continuity correction

data: qr_12_50 and spqr_12_50
V = 37881, p-value < 2.2e-16
alternative hypothesis: true location shift is not equal to 0

```
# QR - SPQR Dec 95  
wilcox.test(x = qr_12_95, y = spqr_12_95, alternative = 'two.sided', paired  
  ↪ = T)
```

Wilcoxon signed rank test with continuity correction

data: qr_12_95 and spqr_12_95

V = 39060, p-value < 2.2e-16

alternative hypothesis: true location shift is not equal to 0

The test results suggest that, for both the median and the 0.95 quantile, the Gaussian, QR and SPQR predictions differ statistically from each other.

Final deliverables

Median p-values:

	Gaussian	QR	SPQR
Gaussian	X	2.2e-16	2.2e-16
QR	2.2e-16	X	2.2e-16
SPQR	2.2e-16	2.2e-16	X

0.95 quantile p-values:

	Gaussian	QR	SPQR
Gaussian	X	2.2e-16	2.2e-16
QR	2.2e-16	X	2.2e-16
SPQR	2.2e-16	2.2e-16	X